CUI TECH FEST · AI WEB FORGE HACKATHON

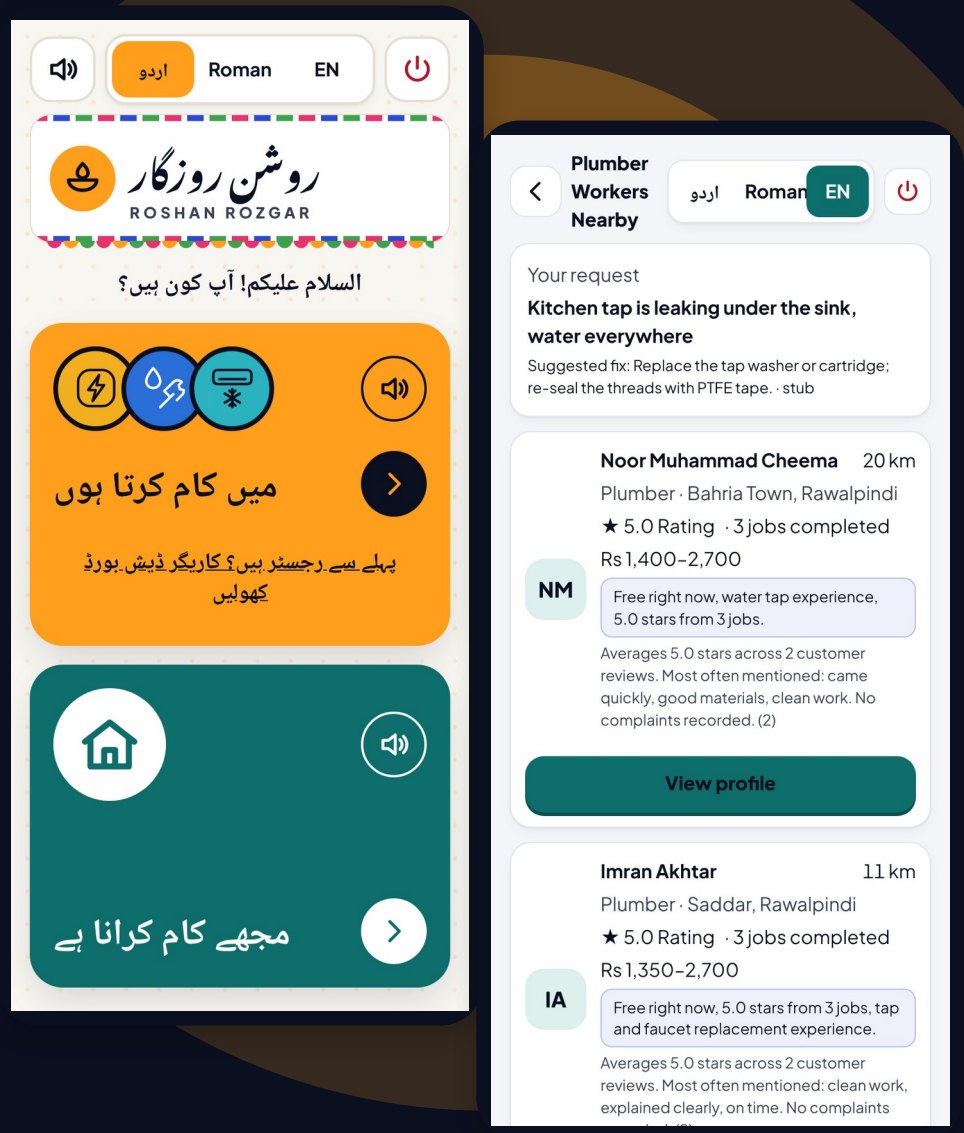
Roshan Rozgar

روشن روزگار

The job platform that doesn't ask a worker to read.

A voice-first, Urdu-first marketplace that connects households with nearby, available, trusted skilled trade workers — electricians, plumbers, AC technicians, carpenters and more.

Team [Team name]
COMSATS University Islamabad · Full-Stack Web Development & Applied AI track



THE PROBLEM

Skilled workers are invisible online

72.1%

of Pakistan's non-agricultural employment is informal

Electricians, plumbers, AC technicians and carpenters are a large share of it. They find work at a street-corner adda, through contractors, or by word of mouth.

Source: Pakistan Bureau of Statistics, Labour Force Survey

The worker

- Every job portal assumes he reads, types into forms and has a CV
- Years of skill, no written proof of it
- Waits at the adda for work that may not come

The household

- Something breaks — who is nearby and free right now?
- Is he competent for this specific job?
- Is he safe to let into my home?

The real barrier is literacy, not language. A form translated into Urdu still needs reading and typing.

OUR SOLUTION

Two skins over one system

Worker skin

Urdu-first · voice-first

- Builds a profile by speaking — no keyboard needed
- Large tap targets, icons + colour for every trade
- Every screen can be read aloud

Customer skin

Fast · dense · trilingual

- Describe the problem by text, voice or photo
- Ranked nearby workers, each with its reason
- Book, agree a price, track live, review

One backend

one database

one booking engine

one AI layer

Speak
Worker profile

→

Describe
AI finds the trade

→

Match
Nearby & free now

→

Agree
Price, both confirm

→

Book
Worker locked: busy

→

Review
Trust for the next

Five outcomes, both roles, one flow



The first screen: “I do work” (amber) or “I need work done” (teal) — chosen by icon and colour, with a speaker button to hear it.

1. Onboarding

Worker

A complete profile from a spoken interview — no form, no typing

2. Discovery

Customer

Describe the problem → nearby, available workers, ranked with reasons

3. Communication

Both

Price offer and two-sided price agreement; chat stores Urdu + English

4. Booking

Both

A state machine; the database refuses a second active job

5. Trust

Customer

Stars, tag chips, written or spoken review → AI trust summary

Every screen switches between اردو · Roman Urdu · English

WHAT WE BUILT

Eight features, each tied to a real barrier

Voice-interview onboarding

Speak name, trade, years, area and skills — AI turns it into a profile

Problem → trade, by AI

Text, voice or photo is classified into trade, urgency and likely fix

Explainable ranking

Availability → distance → skill → rating, with the reason on every card

Trilingual + read-aloud

Urdu, Roman Urdu and English on every screen; text-to-speech

Price offer & agreement

Customer offers, worker proposes, both confirm a structured price

No double-booking

Transactional state machine + a partial unique index in Postgres

Reviews → trust summary

Stars, tag chips, review text; AI summary shows its source count

Honest AI labels

“AI” chip only when a real model answered; “demo data” otherwise

A complete profile, built by speaking



A four-step voice interview replaces the sign-up form. The worker answers aloud in Urdu or Roman Urdu; AI extracts name, trade, years, area, skills and rate band.

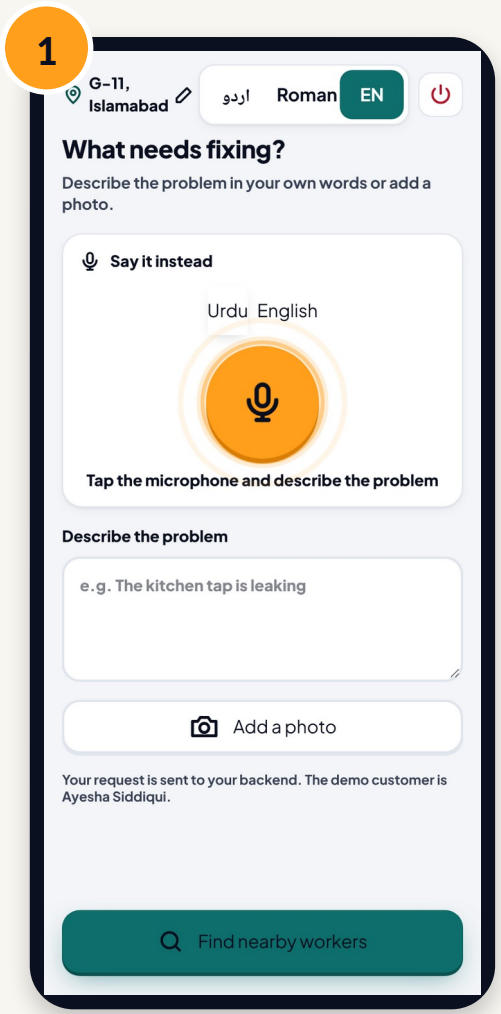
1. **He speaks**
   “Mera naam Bilal Ahmed hai, main plumber hoon, 8 saal…” → transcript appears, fields fill in

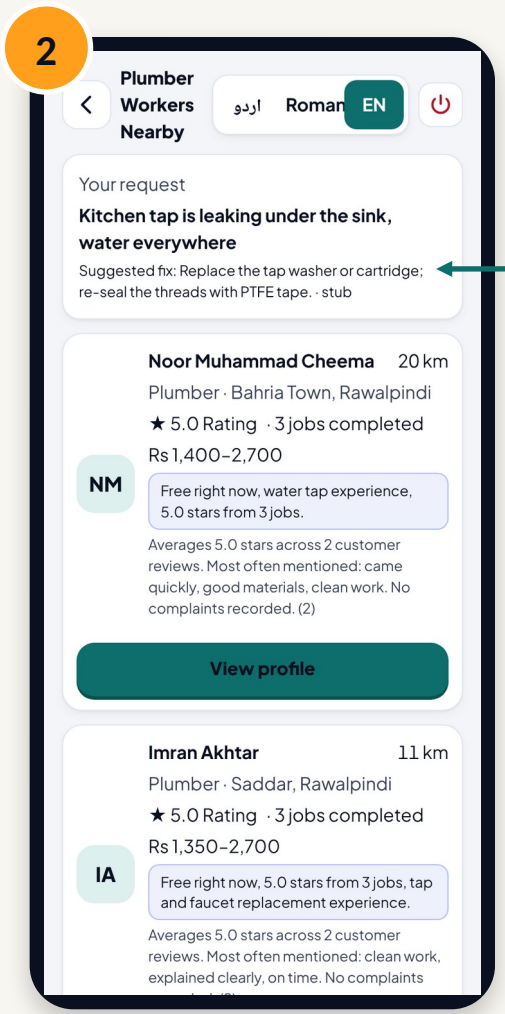
2. **Trade is pre-selected**
   “Are you a plumber?” — confirm with one tap on an icon tile, or say another trade

3. **Skills from his own words**
   8 years, tap and geyser skills quoted back; step 4 adds work photos, reads the profile aloud and publishes

Benefit: no keyboard, no CV — the only way this user gets on.

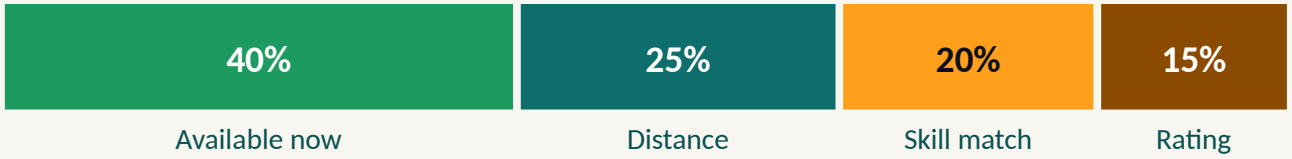
Say what's broken — get who's free, nearby, and why





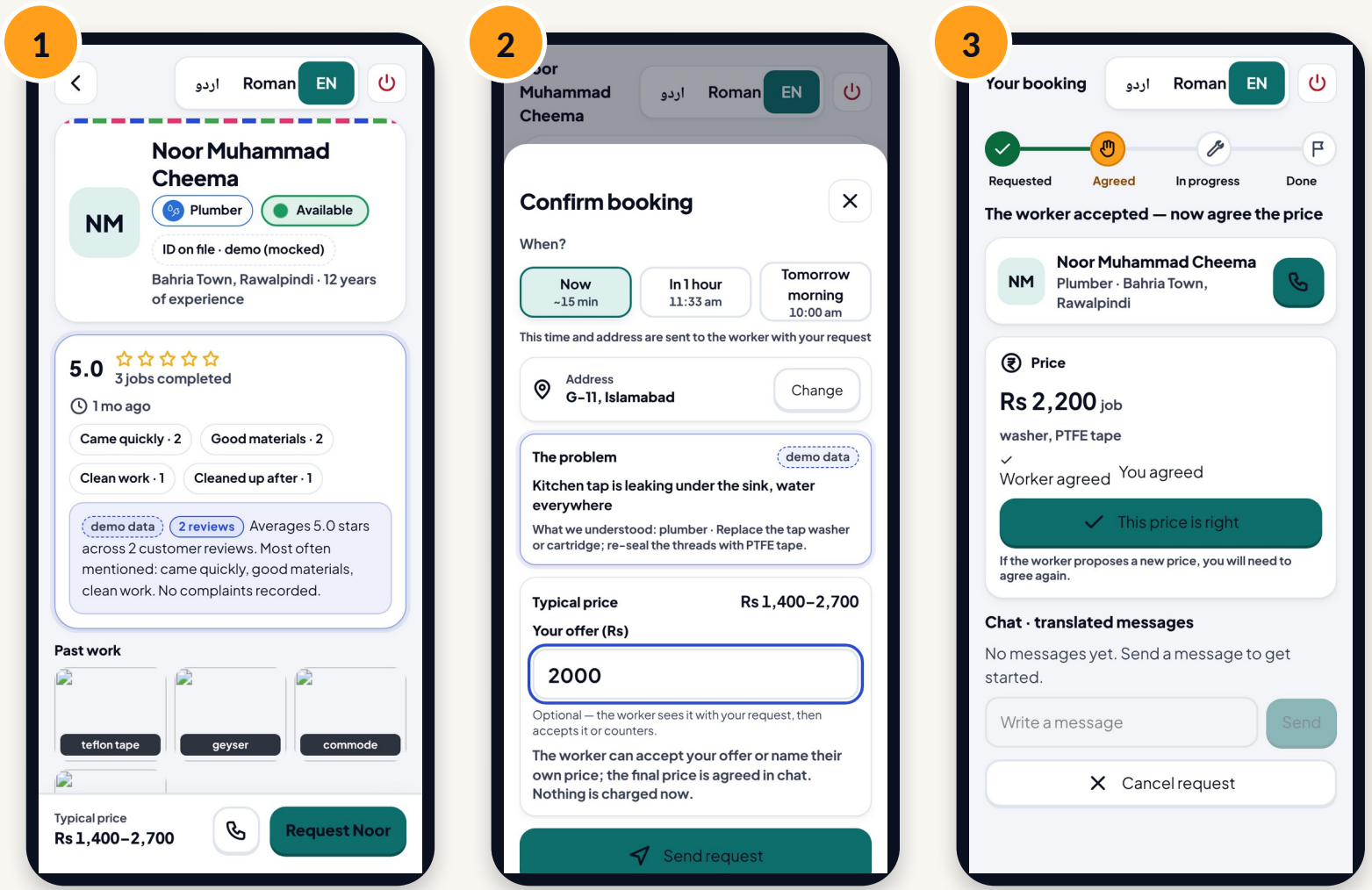
The reason, on every card
"Free right now, water tap experience, 5.0 stars from 3 jobs."

How workers are ranked



Input: text, voice or photo — AI returns trade, urgency, likely fix and keywords.
Benefit: she doesn't need to know the trade's name, and can see why each worker is shown.

From profile to an agreed price in three taps



1 Check the worker
Trade, area, years, "Available" badge, star rating, top tags and an AI trust summary with its review count

2 Request + optional offer
Choose Now / 1 hour / tomorrow, confirm the address, offer a price (here Rs 2,000)

3 Both sides agree
Worker proposes Rs 2,200 incl. washer and PTFE tape → customer taps "This price is right"

Benefit: a structured price both sides confirmed — a record instead of a roadside argument. Chat stores every message in Urdu and English.

No double-booking — the database says no

REQUESTED → ACCEPTED → PRICE_AGREED → IN_PROGRESS → COMPLETED

Side exits: DECLINED · CANCELLED

Active job → worker is BUSY and hidden from “available now”

```
db/migrations/0000_*.sql

CREATE UNIQUE INDEX one_active_job_per_worker
  ON jobs (worker_id)
  WHERE status IN ('ACCEPTED',
                   'PRICE_AGREED', 'IN_PROGRESS');
```

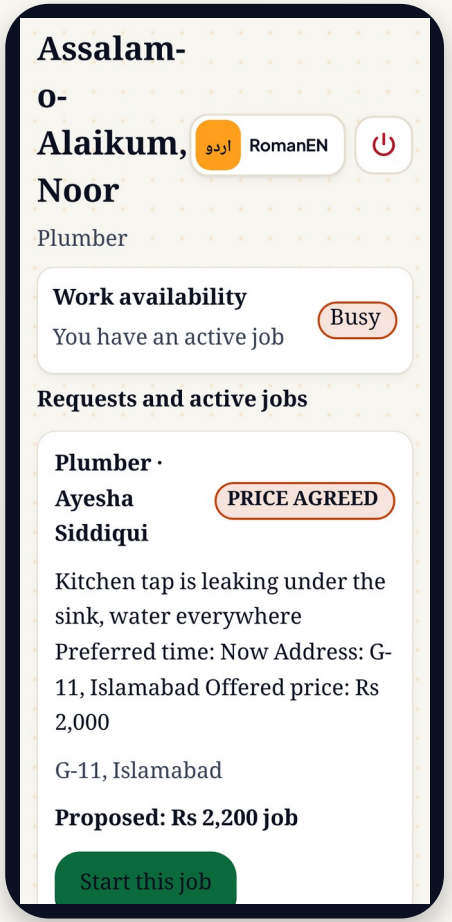
Accept = one transaction: job → ACCEPTED, worker → BUSY, the worker’s other pending requests auto-declined. Complete flips him back to FREE and recounts his jobs.

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database invariant checks pass (npm run verify)

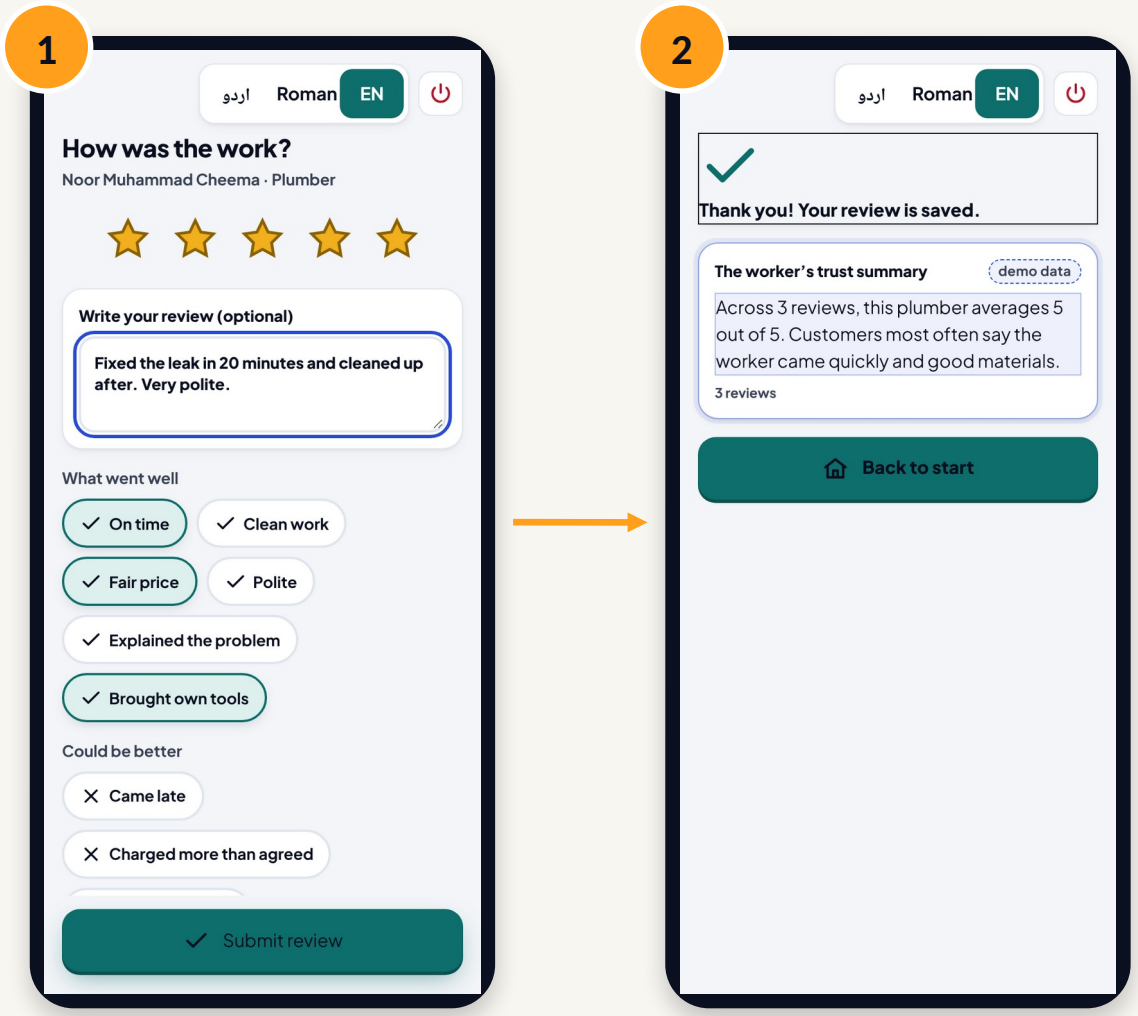
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race-test checks pass, incl. double-accept (npm run race)



Worker home after price agreed: “Busy”

Every finished job makes the next choice easier



1. **Rate in seconds**
   5 stars, tap chips like “On time”, “Fair price”, “Brought own tools”; optional written or spoken review

2. **Trust summary regenerates**
   The worker’s summary is rebuilt from all his reviews and shows its source: 2 reviews → 3 reviews

3. **Rating stays true**
   Rating is recomputed on every review; an invariant check confirms rating = real average

Benefit: the next customer gets a summary of real experiences instead of “ask a neighbour”.

HOW IT WORKS

From a spoken sentence to a finished, reviewed job

	1 Join	2 Ask	3 Match	4 Agree & book	5 Finish & trust
Worker	Speaks his profile; AI extracts it; he confirms and publishes	Goes “available”	Sees the request; accepts with one tap → BUSY	Proposes a price; starts the job	Completes → FREE again; jobs count +1
	One shared Job record in Postgres — both screens poll it every 4–5 s, so they always show the same state				
Customer	Picks area or live location	Describes problem: text, voice or photo → AI finds trade	Gets free workers nearby, ranked, with reasons	Requests + offers a price; confirms worker’s price; tracks live	Rates, tags, reviews → trust summary updates

TECHNOLOGY STACK

Chosen for reliability over novelty

Frontend

Next.js 16 · React 19
App Router; both skins in one app

TypeScript (strict)
One type system, front to back

Browser speech + mic APIs
Recording and read-aloud fallback

Backend

Next.js Route Handlers
29 API routes, one deploy, no CORS

Zod 4
Validation at every boundary

Job state machine
Transactional transitions

Data & storage

PostgreSQL (Supabase)
Partial unique index, jsonb, enums

PGlite fallback
Embedded Postgres — runs with no cloud

Drizzle ORM · Supabase Storage
Migrations; audio & photo uploads

AI services

Claude (Anthropic) · Gemini
Extract, classify, tag photos, translate, price, summarise

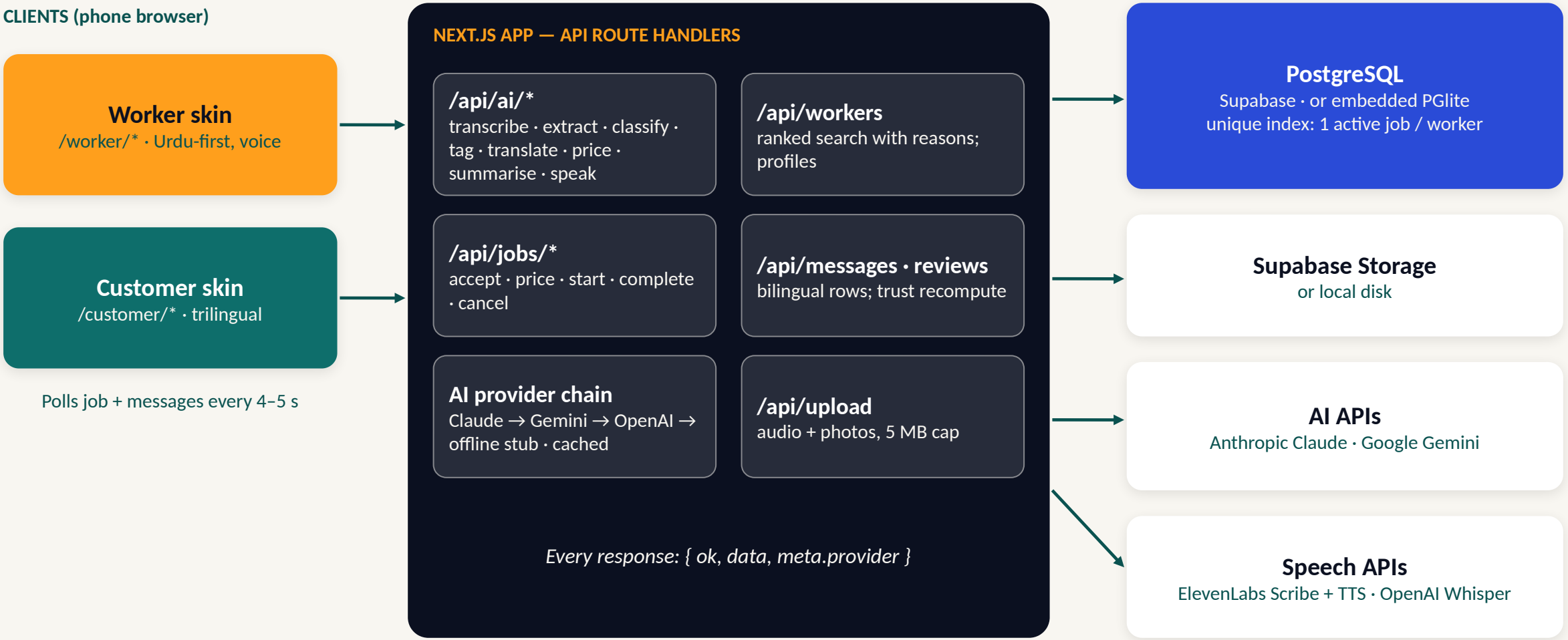
ElevenLabs Scribe · Whisper
Urdu speech-to-text

ElevenLabs eleven_v3
Urdu text-to-speech

Hosting: Vercel, plus a local instance as demo backup. **Built with:** Claude Code as AI coding assistant (disclosed in README).

SYSTEM ARCHITECTURE

One app, one database, a fail-safe AI layer



Why a worker — and a household — would choose this

	Typical job portal / classifieds	Roshan Rozgar
Joining	Typed form, CV upload	**A spoken interview in Urdu**
Finding help	Pick the right category yourself	**Describe it in your words, voice or photo**
Ranking	Opaque order	**A plain-language reason on every card**
Availability	Stale listings	**Free / busy enforced by the database**
Trust	A star count	**AI summary with its source count**

Also built for venue reality: runs with no keys and no cloud (PGlite + offline stub), and labels it honestly.

Remove the AI and…

Speech-to-text
the worker cannot onboard at all

Profile extraction
raw audio is not a searchable profile

Problem classification
she must diagnose her own fault first

Translation
the two users cannot negotiate

Review synthesis
trust goes back to “ask a neighbour”

What works today, what's next, and why it matters

Working today

- Voice onboarding (4 steps) → published profile
- Text / voice / photo problem description
- Ranked search with reasons
- Booking, price offer & agreement, live status
- Reviews + trust summary
- Text chat stored in Urdu + English

Being finished

- Worker "incoming request" & "job done" screens are design mock-ups
- Spoken price quote → structured price
- Voice notes inside chat (text today)
- Real-mic test on the demo device
- Login is a mocked phone + PIN

Future scope

- Offline-first job queue for 3G dead zones
- Adda-level vouching: workers vouch for new workers
- Real OTP and ID verification
- Cash-free payments
- More cities beyond Islamabad / Rawalpindi

Impact: brings skilled workers online without asking them to learn to read first — and gives households a faster, safer way to get things fixed.

THANK YOU · شکریہ

Five outcomes. Both roles. One flow.

Roshan Rozgar — the job platform that doesn't ask a worker to read.

A worker speaks once and becomes discoverable. A household describes a problem and gets someone nearby, free and trusted. The database keeps both sides honest.

Team [Team name]
Live demo next — worker on the left, customer on the right

